Revisions

Rev.	Date	Author	Description
A	16/11/2020	Fabrice Pichard	Created from xZV IND-E (Engine Zapi encoder controler) Adapted for retro-compatibility with Sevcon equiped Commuters
B	18/01/2021	FPI	Rectified diode polarity, detailed twisting & shielding, added Canbus separation connector
C	20/01/2021	FPI	Added diode fitting instructions. 24V wire from grey to violet.

D1_Cathode_Splice

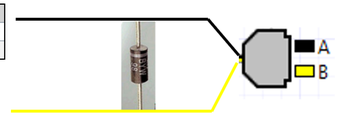
From	To	Conductor	Color	Gauge
D1_Cathode_Splice	Brake_M.A	W34.Violet		1 mm²
D1_Cathode_Splice	D1.NEG	W35.Violet		1 mm²
D1_Cathode_Splice	Power_supply_can_bus.E	W12.Violet		1 mm²

D1_Anode_Splice

From	To	Conductor	Color	Gauge
D1_Anode_Splice	Brake_M.B	W36.Yellow		1 mm²
D1_Anode_Splice	D1.POS	W17.Yellow		1 mm²
D1_Anode_Splice	Variator.K3	W14.Yellow		1 mm²

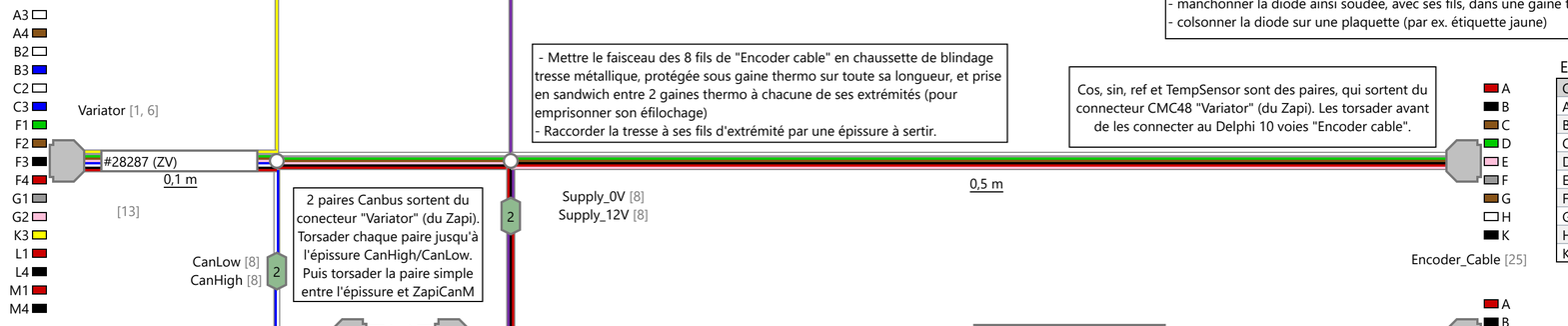
D1

From	To	Conductor	Color	Gauge
D1.NEG	D1_Cathode_Splice	W35.Violet		1 mm²
D1.POS	D1_Anode_Splice	W17.Yellow		1 mm²









- souder chaque patte de la diode à son fil.
- manchonner la soudure par une petite gaine thermo.
- manchonner la diode ainsi soudée, avec ses fils, dans une gaine thermo.
- colsonner la diode sur une plaquette (par ex. étiquette jaune)

- Mettre le faisceau des 8 fils de "Encoder cable" en chaussette de blindage tresse métallique, protégée sous gaine thermo sur toute sa longueur, et prise en sandwich entre 2 gaines thermo à chacune de ses extrémités (pour emprisonner son éfilochage)
- Raccorder la tresse à ses fils d'extrémité par une épissure à sertir.

Cos, sin, ref et TempSensor sont des paires, qui sortent du connecteur CMC48 "Variator" (du Zapi). Les torsader avant de les connecter au Delphi 10 voies "Encoder cable".

2 paires Canbus sortent du conecteur "Variator" (du Zapi). Torsader chaque paire jusqu'à l'épissure CanHigh/CanLow. Puis torsader la paire simple entre l'épissure et ZapiCanM

Replier le fil dénudé sur lui-même pour doubler son épaisseur dans le sertissage de la cosse

Brake_M

Cavity	To	Conductor	Color	Gauge	From Contact PN
A	D1_Cathode_Splice	W34.Violet		1 mm²	
B	D1_Anode_Splice	W36.Yellow		1 mm²	

Encoder_Cable

Cavity	To	Color	Gauge	Conductor	Length
A	Variator.F4		0,25 mm²	W26.Red	≥ 0,6 m
B	Variator.F3		0,25 mm²	W25.Black	≥ 0,6 m
C	Variator.F2		0,25 mm²	W24.Brown	≥ 0,6 m
D	Variator.F1		0,25 mm²	W22.Green	≥ 0,6 m
E	Variator.G2		0,25 mm²	W28.Pink	≥ 0,6 m
F	Variator.G1		0,25 mm²	W27.Gray	≥ 0,6 m
G	Variator.A4		0,5 mm²	W30.Brown	≥ 0,6 m
H	Variator.A3		0,5 mm²	W29.White	≥ 0,6 m
K	Supply_0V		1 mm²	W4.Black	≥ 0,5 m

Codeur Check Mesurement, TEST BEFORE START :
Ref (Pink+/ Grey-) : 80 Ohm
Cos (Red+/ Black-) : 70 Ohm
Sin (Brown+/ Green-) : 70 Ohm
Brake : a few ohms
TempSensor (Brown+/White-) : if KTY 500 Ohm, if PT1000 1000 ohms

Variator

Cavity	To	Color	Gauge	Conductor	Length	Twisted With
A3	Encoder_Cable.H		0,5 mm²	W29.White	≥ 0,6 m	W30
A4	Encoder_Cable.G		0,5 mm²	W30.Brown	≥ 0,6 m	W29
B2	CanHigh		0,5 mm²	W11.White	≥ 0,1 m	W15
B3	CanLow		0,5 mm²	W15.Blue	≥ 0,1 m	W11
C2	CanHigh		0,5 mm²	W7.White	≥ 0,1 m	W6
C3	CanLow		0,5 mm²	W6.Blue	≥ 0,1 m	W7
F1	Encoder_Cable.D		0,25 mm²	W22.Green	≥ 0,6 m	W24
F2	Encoder_Cable.C		0,25 mm²	W24.Brown	≥ 0,6 m	W22
F3	Encoder_Cable.B		0,25 mm²	W25.Black	≥ 0,6 m	W26
F4	Encoder_Cable.A		0,25 mm²	W26.Red	≥ 0,6 m	W25
G1	Encoder_Cable.F		0,25 mm²	W27.Gray	≥ 0,6 m	W28
G2	Encoder_Cable.E		0,25 mm²	W28.Pink	≥ 0,6 m	W27
K3	D1_Anode_Splice		1 mm²	W14.Yellow	≥ 0,1 m	
L1	Supply_12V		0,5 mm²	W8.Red	≥ 0,1 m	
L4	Supply_0V		0,5 mm²	W13.Black	≥ 0,1 m	
M1	Supply_12V		0,5 mm²	W10.Red	≥ 0,1 m	
M4	Supply_0V		0,5 mm²	W3.Black	≥ 0,1 m	

ZapiCanF

From	To	Conductor	Color	Gauge	From Contact PN	Twisted With
ZapiCanF.A	Power_supply_can_bus.C	W5.White		1 mm²		W31
ZapiCanF.B	Power_supply_can_bus.D	W31.Blue		1 mm²		W5

ZapiCanM

From	To	Conductor	Color	Gauge	From Contact PN	Twisted With
ZapiCanM.A	CanHigh	W9.White		1 mm²		W16
ZapiCanM.B	CanLow	W16.Blue		1 mm²		W9

Power_supply_can_bus

Cavity	To	Conductor	Color	Gauge	From Contact PN	Twisted With
A	Supply_12V	W2.Red		1 mm²	Delphi_150_contact_male	
B	Supply_0V	W1.Black		1 mm²	Delphi_150_contact_male	
C	ZapiCanF.A	W5.White		1 mm²	Delphi_150_contact_male	W31
D	ZapiCanF.B	W31.Blue		1 mm²	Delphi_150_contact_male	W5
E	D1_Cathode_Splice	W12.Violet		1 mm²		

Supply_0V, Supply_12V

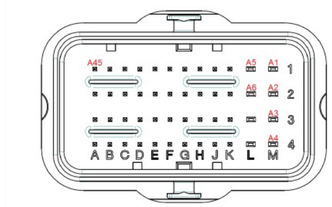
From	To	Conductor	Color	Gauge
Supply_0V	Encoder_Cable.K	W4.Black		1 mm²
Supply_0V	Power_supply_can_bus.B	W1.Black		1 mm²
Supply_0V	Variator.L4	W13.Black		0,5 mm²
Supply_0V	Variator.M4	W3.Black		0,5 mm²
Supply_12V	Power_supply_can_bus.A	W2.Red		1 mm²
Supply_12V	Variator.L1	W8.Red		0,5 mm²
Supply_12V	Variator.M1	W10.Red		0,5 mm²

CanHigh, CanLow

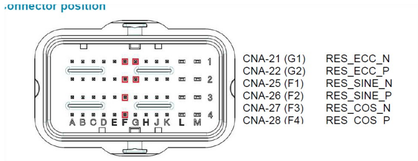
From	To	Conductor	Color	Gauge
CanHigh	Variator.B2	W11.White		0,5 mm²
CanHigh	Variator.C2	W7.White		0,5 mm²
CanHigh	ZapiCanM.A	W9.White		1 mm²
CanLow	Variator.B3	W15.Blue		0,5 mm²
CanLow	Variator.C3	W6.Blue		0,5 mm²
CanLow	ZapiCanM.B	W16.Blue		1 mm²

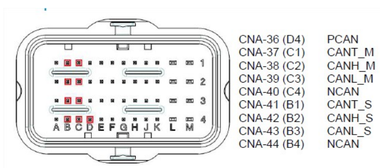
Low voltage connector (Molex CNC 48 poles)

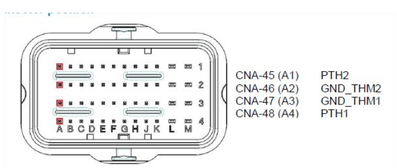
Low voltage signals are connected to the HV Inverter through a 48 pins Molex CNC connector

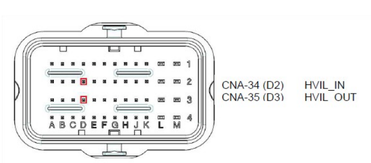


Connector position









Pin	Coor.	Type	Name	Description
A1	M1	Input	KEY	Input of the key switch signal (low voltage supply positive 12V or 24V)
A2	M2	Input	PEV	Connect the positive supply of electrovalves (EB, EVP and EVx) to this pin. Take the positive supply immediately after the main contactor.
A3	M3	Output	PANIN	Positive supply for external analog device (Sin/Cos sensor, potentiometers, ecc) factory selectable +12/+5V, maximum 100mA
A4	M4	Input	N_KEY	Low voltage power supply negative
A5	L1	Input	KEY	Input of the key switch signal (low voltage supply positive 12V or 24V)
A6	L2	Input	PEV	Connect the positive supply of electrovalves (EB, EVP and EVx) to this pin. Take the positive supply immediately after the main contactor.
A7	L3	Output	AGND	Ground for external analog devices (Sin/Cos sensor, potentiometers, ecc)
A8	L4	Input	N_KEY	Low voltage power supply negative

Bill of Materials

Id	Type	Manufacturer	Part Number	Description	Quantity
1	Boot	Molex	Connecteurs automobile CMC WIRE CAP FOR 28P AND 48P	538-64320-1301	1
2	Cable		2 Wire (1 twisted pair), 20 AWG Unshielded	Generic Cable	0 in
3	Cable	LAPP	#20811 - Câble vert OLFLEX (4x2x0.25mm² + 2x0.5mm²) LAP0036927		0 in
6	Connector	Molex	#27627- Connecteur Molex 48 Pin	538-64320-1311	1
7	Contact	Delphi	Delphi_150_contact_male		4
8	Splice		Generic Splice	Generic Splice	6
9	Wire		0.5mm² Black		≥ 0,2 m
10	Wire		0.5mm² Red		≥ 0,2 m
11	Wire		1mm² Black		≥ 0,5 m
12	Wire		1mm² Red		
13	Label		Generic White Label	Generic Label	3
14	Wire		0.5 mm² - Blue		≥ 0,2 m
15	Wire		0.5 mm² - White		≥ 0,8 m
16	Wire		1 mm² Violet	Generic Wire	
17	Connector	Delphi	#16777 - Delphi_280_2_way_male	829-15300002	1
18	Wire		1 mm² Yellow	Generic Wire	≥ 0,1 m
19	Wire		1mm² - White		
20	Wire		1mm² - Blue		
21	Connector		#14017 - Delphi_150_6_way_male	829-12124107	1
22	Wire		0.75mm2 yellow		0 in
23	Diode		Diode traversante, 5A, 60V, SB560-E3/54	#29732	1
24	Wire		Grey - red stripe 1 mm2		0 in
25	Connector	Delphi	#22965 - Delphi_150_10_ways_female	829-12065425	1
26	Wire		0.25² Green		≥ 0,6 m
27	Wire		brown		≥ 0,6 m
28	Wire		0.25² black		≥ 0,6 m
29	Wire		0.25² red		≥ 0,6 m
30	Wire		0.25² grey		≥ 0,6 m
31	Wire		0.25² pink		≥ 0,6 m
32	Wire		0.5 mm² Brown	Generic Wire	≥ 0,6 m
33	Connector	Delphi	#14869 - Delphi_150_2_way_female	829-12052641	1
34	Connector	Delphi	#15215 - Delphi_150_2_way_male	829-12162000	1
35	Wire		drain		0 in
36	Wire		drain		0 in

	Name	Date
Drawn By	----	----
Eng Approved	----	----
Mfg Approved	----	----
QA	----	----

Title: #31314 - Engine_zapi_encoder_controler (ZV) - A2-IND-C

Description:

Revision: ----

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